A variance bound for transverse rhythmic triplets and monochromatic queens

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Abstract

We give a short second-moment proof that a same-color nonattacking placement of t queens on a $k \times k$ board satisfies $t^2 - 1 \le 3k(k-t)$. Equivalently, this bounds a rainbow packing of three-term arithmetic progressions that each meet three consecutive length-k blocks once. The proof and its application to actual exact-cover rhythmic tilings have been checked in Lean 4. The result is a supporting obstruction: it does not establish existence of perfect rhythmic triplet tilings for all sufficiently large orders. Novelty remains unestablished.

1 Statement and relation to rhythmic tilings

A perfect rhythmic triplet tiling of order k partitions $\{0, \dots, 3k-1\}$ into k sets $\{a_i, a_i + d_i, a_i + 2d_i\}$ with positive, pairwise distinct integer gaps d_i. Translation by 1 gives the equivalent convention $\{1, \dots, 3k\}$. A triplet is *transverse* if it meets each consecutive length-k block once.

Theorem 1 (Partial transverse packing bound)**.** *Let $a, b, c : \{1, \dots, t\} \to \mathbb{Z}$ be individually injective, with values in $\{0, \dots, k-1\}$, and suppose*

$$a_i + c_i = 2b_i, \qquad a_i - c_i \neq a_j - c_j \quad (i \neq j).$$

Then

$$t^2 - 1 \le 3k(k-t). \tag{1}$$

Consequently (1) holds for the number t of transverse triplets in any perfect rhythmic triplet tiling of order k.

The core statement does not assume that the partial packing extends to a perfect tiling. To pass from a tiling to this statement, subtract k from every transverse middle term and $2k$ from every transverse last term. Exact coverage gives the three injective coordinate lists; the original gap is $k + (c_i - a_i)/2$.

Theorem 2 (One-color queens bound)**.** *Let $(a_i, c_i) \in \{0, \dots, k-1\}^2$, $1 \le i \le t$, be mutually nonattacking queens, all on the same checkerboard color. Thus the four lists $a_i, c_i, a_i + c_i, a_i - c_i$ are individually injective, and $a_i + c_i$ has a common parity. Then (1) holds.*

*AI research project author label. This note and its Lean sources were generated with AI assistance. No claim of established novelty is made.

For even $a_i + c_i$, set $b_i = (a_i + c_i)/2$. This is a bijective correspondence between these queen placements and partial transverse rainbow triplet packings, via

$$(a_i, c_i) \longleftrightarrow (a_i,\ k + (a_i + c_i)/2,\ 2k + c_i).$$

Rows and columns encode endpoint disjointness; the two diagonal directions encode distinct middle terms and gaps. These are partial packings, not necessarily completable tilings. The queens theorem also covers odd parity directly.

2 Proof by second moments

Lemma 3 (Spaced-list variance)**.** *If $z_1 < \cdots < z_m$ are real numbers with $z_{i+1} - z_i \geq q \geq 0$, then*

$$12 \sum_{i=1}^{m} z_i^2 \ \geq\ q^2 m(m^2 - 1).$$

Proof. The assertion is immediate for $m = 0$. Otherwise,

$$\begin{aligned} m \sum_i z_i^2 - \left(\sum_i z_i \right)^2 &= \sum_{i<j} (z_j - z_i)^2 \\ &\geq q^2 \sum_{i<j} (j - i)^2 = \frac{q^2 m^2 (m^2 - 1)}{12}. \end{aligned}$$

Discard the nonnegative square and divide by m. □

Proof of Theorem 2. Injectivity of the row coordinates gives $t \leq k$; write $r = k - t$ and $h = (k-1)/2$. Let

$$X_i = a_i - h, \quad Y_i = c_i - h, \quad U_i = X_i + Y_i, \quad V_i = X_i - Y_i.$$

The U_i are distinct and have pairwise differences in $2\mathbb{Z}$, and the same is true of the V_i. Applying Lemma 3 with spacing 2 after sorting gives

$$\sum_i U_i^2 + \sum_i V_i^2 \geq \frac{2t(t^2 - 1)}{3}. \tag{2}$$

The full centered coordinate list has square sum

$$\sum_{j=0}^{k-1} (j - h)^2 = \frac{k(k^2 - 1)}{12}.$$

The r omitted row coordinates are distinct integers translated by h, so Lemma 3, with spacing 1, bounds their square sum below by $r(r^2 - 1)/12$. Therefore

$$\sum_i X_i^2 \leq \frac{k(k^2 - 1) - r(r^2 - 1)}{12},$$

and the same bound holds for $\sum_i Y_i^2$. Since $U_i^2 + V_i^2 = 2(X_i^2 + Y_i^2)$, combining with (2) yields

$$2t(t^2 - 1) \leq k(k^2 - 1) - r(r^2 - 1).$$

If $t > 0$, substitute $r = k - t$, factor out t, and simplify to (1). If $t = 0$, the required inequality is $-1 \leq 3k^2$. □

Theorem 1 follows from the even-parity queens correspondence, and its tiling consequence follows from the coordinate restriction described above.

Corollary 4 (Asymptotic consequence)**.** *For $k > 0$,*

$$t \le \frac{\sqrt{21k^2 + 4} - 3k}{2}.$$

In particular, the limiting upper density of such queen placements or transverse partial packings is at most

$$\frac{\sqrt{21} - 3}{2} = 0.7912878474\ldots.$$

Every sequence of perfect rhythmic triplet tilings with orders tending to infinity therefore has non-transverse proportion with lower limit at least

$$\frac{5 - \sqrt{21}}{2} = 0.2087121525\ldots.$$

Proof. Solve $t^2 + 3kt - 3k^2 - 1 \le 0$ for the nonnegative root, divide by k, and take the limit. For the tiling statement the proportion is $1 - t/k$. □

3 Formal verification and scope

The accompanying sources were compiled with Lean 4.32.2 and Mathlib commit

`905b95818eb32af7874a58b427f50c1711a5e96c`.

The final theorems contain no assumed variance bound or assumed packing estimate:

- `Rhythmic.partial_transverse_bound`: the actual injective-array bound. The formal core needs bounds only on a, c; midpoint bounds follow in the unshifted case.
- `Rhythmic.PerfectTripletTiling.transverse_bound`: the number of transverse triplets in a genuine exact-cover tiling.
- `Rhythmic.monochromatic_queens_bound`: the bound from bounded integer row/column coordinates, four injectivity conditions, and a common sum remainder modulo 2.

An exact-cover tiling is represented by positive distinct gaps together with a bijection

$$\mathrm{Fin}(k) \times \mathrm{Fin}(3) \simeq \mathrm{Fin}(3k)$$

whose value is the specified start plus the term index times the gap. Thus the formal tiling theorem concerns an actual partition, rather than merely a numerical condition.

All inequalities are stated using exact casts into $\mathbb{R}$, with the subtraction $k - t$ taken there rather than truncated natural-number subtraction. The displayed integer inequality is equivalent. The square-root reformulation and limiting constants above are mathematical corollaries; they are not separately formalized. Lean's axiom audit reports only `propext`, `Classical.choice`, and `Quot.sound`; there are no proof placeholders or added axioms. An independent fresh source rebuild also passed, with matching source hashes and exact theorem-type inspection. The package includes source files, a rebuild script, and both compiler and independent-audit receipts.

4 Prior art and limitations

The exact eventual-existence question for perfect rhythmic triplet tilings is explicitly left open in the 2026 packing papers [1, 2]. The July paper establishes an asymptotic prescribed-gap packing result, not an exact tiling theorem. Johnson's original account gives examples and elementary restrictions [3]; tiling counts are catalogued in OEIS A261516 [4].

The partial transverse extremal problem is already the one-color queens problem in OEIS A352241 [5]. Its linked 2022 discussion [6] contains computational counterexamples to a proposed bound. A 2025 discussion [7] gives an obstruction to placing $k-1$ same-color queens on larger boards and constructions with asymptotic density $2/3$; it does not state the bound proved here. A targeted search found no instance of (1), but that does not establish novelty. In particular, the full text of Davalan's substantial treatment [8] was not available for review, and a linked Greek forum discussion could not be retrieved. The variance argument is elementary enough that an equivalent result may be known under other terminology.

Unresolved main objective. Nothing here proves that every sufficiently large k admits a perfect rhythmic triplet tiling. Computational witnesses, unsuccessful construction searches, and specialized inflation obstructions remain research checkpoints, not substitutes for that existence theorem. No publication or external submission has been made.

References

[1] N. Alon, M. Dębski, J. Grytczuk, and J. Przybyło, *Packing arithmetic progressions*, 2026. arXiv:2603.02786v1.

[2] Y. Mao, Z. Wang, M. Wei, and G. Yang, *Solution to a conjecture of Alon, Dębski, Grytczuk and Przybyło on fixed-cardinality arithmetic progressions*, 2026. arXiv:2607.06113v1.

[3] T. Johnson, *Perfect Rhythmic Tilings*, MaMuX lecture, IRCAM, 24 January 2004. Author's lecture text.

[4] OEIS Foundation, A261516: Perfect rhythmic tilings with triples.

[5] OEIS Foundation, A352241: Maximal number of nonattacking black-square queens.

[6] G. Baloglou and contributors, *Black queens on an $n \times n$ board*, Mathematics Stack Exchange, March 2022. Original discussion.

[7] H. Reddmann, daw, AxiomaticSystem, and contributors, *8 queen puzzle unicolor variant*, Puzzling Stack Exchange, December 2025. Original discussion.

[8] J.-P. Davalan, *Perfect Rhythmic Tilings*, Perspectives of New Music **49**(2) (2011), 144–197. doi:10.7757/persnewmusi.49.2.0144. Bibliographic record checked; full text not reviewed.